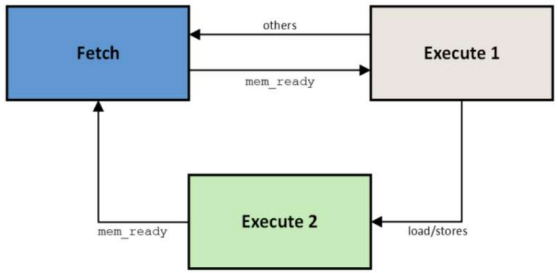
Sel_nstate

State	opcode	mem_ready	Next state
Fetch	x	1'b0	Fetch
		1'b1	Exec1
Exec1	R_to_R	x	Fetch
	Imm	x	Fetch
	LUI	x	Fetch
	AUIPC	x	Fetch
	Loads	x	Exec2
	Stores	x	Exec2
	Cond_b	x	Fetch
	JAL	x	Fetch
	JALR	x	Fetch
Exec2	x	1'b0	Exec2
		1'b1	Fetch



Memory and Reg_Ctrl

	Output signals					
State	selPC_addr	selALU_addr	read_en	selINC_PC	loadIR	loadPC
Fetch	1'b1		1'b1		mem_ready	
Exec1				opcode != Loads && opcode != Stores && opcode != JAL && opcode != JALR && opcode != Cond_b		opcode != Loads && opcode != Stores
Exec2		1'b1	opcode == Loads	1'b1		mem_ready

Encoder_flag

	Output signals					
State	Type_R	Type_I	Type_S	Type_SB	Type_U	Type_UJ
Fetch	opcode == R_to_R	opcode == Imm || opcode == Loads || opcode == JALR	opcode == Stores	opcode == Cond_b	opcode == LUI || opcode == AUIPC	opcode == JAL
Exec1						
Exec2						

ALU

			Output signals													
State	opcode	{func7,func3}	ALU_op	ldALU	ldMEM	ldPC	ce	we_n	selRS1	selRS2	selImm	selPC	selImm_PC	selJump	selZero_cond	selNZero_cond
Fetch																
Exec1	R_to_R	{7'b0000000, 3'b000}	ADD	1'b1			1'b1	1'b0	1'b1	1'b1						
		{7'b0100000, 3'b000}	SUB	1'b1			1'b1	1'b0	1'b1	1'b1						
		{7'b0000000, 3'b010}	SLT	1'b1			1'b1	1'b0	1'b1	1'b1						
		{7'b0000000, 3'b011}	SLTU	1'b1			1'b1	1'b0	1'b1	1'b1						
		{7'b0000000, 3'b100}	XOR	1'b1			1'b1	1'b0	1'b1	1'b1						
		{7'b0000000, 3'b001}	SLL	1'b1			1'b1	1'b0	1'b1	1'b1						
		{7'b0000000, 3'b101}	SRL	1'b1			1'b1	1'b0	1'b1	1'b1						
		{7'b0100000, 3'b101}	SRA	1'b1			1'b1	1'b0	1'b1	1'b1						
		{7'b0000000, 3'b110}	OR	1'b1			1'b1	1'b0	1'b1	1'b1						
		{7'b0000000, 3'b111}	AND	1'b1			1'b1	1'b0	1'b1	1'b1						
	Imm	{7'b???????, 3'b000}	ADD	1'b1			1'b1	1'b0	1'b1		1'b1					
		{7'b???????, 3'b010}	SLT	1'b1			1'b1	1'b0	1'b1		1'b1					
		{7'b???????, 3'b011}	SLTU	1'b1			1'b1	1'b0	1'b1		1'b1					
		{7'b???????, 3'b100}	XOR	1'b1			1'b1	1'b0	1'b1		1'b1					
		{7'b???????, 3'b110}	OR	1'b1			1'b1	1'b0	1'b1		1'b1					
		{7'b???????, 3'b111}	AND	1'b1			1'b1	1'b0	1'b1		1'b1					
		{7'b0000000, 3'b001}	SLL	1'b1			1'b1	1'b0	1'b1		1'b1					
		{7'b0000000, 3'b101}	SRL	1'b1			1'b1	1'b0	1'b1		1'b1					
		{7'b0100000, 3'b101}	SRA	1'b1			1'b1	1'b0	1'b1		1'b1					
	LUI	x	PASS_B	1'b1			1'b1	1'b0			1'b1					
	AUIPC	x	ADD	1'b1			1'b1	1'b0			1'b1	1'b1				
	Loads	x	ADD				1'b1		1'b1		1'b1					
	Stores	x	ADD				1'b1		1'b1		1'b1					
	Cond_b	{7'b???????, 3'b000}	XOR				1'b1		1'b1	1'b1					1'b1	
		{7'b???????, 3'b001}	XOR				1'b1		1'b1	1'b1						1'b1
		{7'b???????, 3'b100}	SLT				1'b1		1'b1	1'b1					1'b1	
		{7'b???????, 3'b101}	SLT				1'b1		1'b1	1'b1						1'b1
		{7'b???????, 3'b110}	SLTU				1'b1		1'b1	1'b1					1'b1	
		{7'b???????, 3'b111}	SLTU				1'b1		1'b1	1'b1						1'b1
	JAL	x				1'b1	1'b1	1'b0					1'b1			
	JALR	x	ADD			1'b1	1'b1	1'b0	1'b1		1'b1			1'b1		
Exec2	Loads	x	ADD		mem_ready		1'b1	~mem_ready	1'b1		1'b1					
	Stores	x	ADD				1'b1		1'b1		1'b1					

Mem_Data_Formatting

			Output signals							
State	Opcode	func3	SgnE_8	SgnE_16	PT	ZrE_8	ZrE_16	ld_SB	ld_SH	ld_SW
Fetch										
Exec1										
Exec2	Loads	3'b000	1'b1							
		3'b001		1'b1						
		3'b010			1'b1					
		3'b100				1'b1				
		3'b101					1'b1			
	Stores	3'b000						1'b1		
		3'b001							1'b1	
		3'b010								1'b1